

The Economics of Energy

Columbia University SIPA

Spring 2026

Instructor: Prof Robert Metcalfe

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Office hours: Thursday, 3-4pm (Room IAB 1432).

Course Description: This course is an introduction to the economics of energy markets. We will study the main sources of inefficiencies in energy markets – market power and externalities – and their implications for policy making. We will cover, for example, how oil and natural gas prices are determined globally, deregulation and market power in electricity markets, and policy responses such as carbon taxes or permits trading. We will also discuss recent challenges faced by policymakers in energy markets, such as the incorporation of renewables, transmission, transport, and the energy transition to net zero more broadly.

Students will work in groups to deliver a project and presentation focusing on a real-world energy policy question of choice. The objective of the group project is to give hands-on experience in energy policy analysis using main concepts of economics discussed in the class.

Meeting dates/times:

• Lecture:

- Section 1—Tuesday, 10am-11:50am, 407 IAB
- Section 2— Tuesday, 3:10pm-5pm, 407 IAB

• Recitation:

- Wednesday, 7:10pm-9pm, TA: Alina Gafanova, 405 IAB [ag4821@columbia.edu]
- Thursday, 8am-9:50am, TA: Junho Choi, 405 IAB [jc5341@columbia.edu]

Office hours:

- Tuesday 5:10-7:10pm, Alina Gafanova [ag4821@columbia.edu]. Please sign up and check for the updated location [following this link](#). Zoom option is available
- Thursday 5:10-7:10pm, Junho Choi [jc5341@columbia.edu]. Please sign up and check for the updated location [following this link](#). Zoom option is available

Grading:

- 10% Attendance.
 - First unauthorized absence = 1%
 - Second unauthorized absence = 2%
 - Third unauthorized absence = 3%
 - Fourth unauthorized absence = 4%
 - Authorized absences include medical issues, capstones, and other cases approved at the instructor's discretion. All absences must be communicated *before* the class begins
- 6% Quizzes (3 x 2% each)
- 25% Group project (full paper and in-class presentation)
- 20% Midterm exam
- 39% Final exam

Quizzes (6%): There will be three quizzes in which students will have to solve exercises based on the economic concepts discussed in lecture and recitation. You will work on the problem sets in groups/teams of 3-4

Group project (25%): Students will work in teams of 2-4 to perform a *detailed* economic analysis of an energy policy question using concepts learned in the class. There will be three deliverables for the group project:

1. Project proposal consisting of at most one-page summary of an energy policy question to be analyzed along with a planned methodology. This is due by the 11th class.
2. Project presentation when the group will present emerging findings and policy recommendations. Presentations should consist of no more than eight slides. Each group will be given 15 minutes to present and 5 minutes for questions and answers at the end. The presentations will be held during the last couple of classes of the semester. This deliverable will count as 30% of the final group project grade.
3. Project final paper should include a discussion of the importance of the topic chosen, methods and data (if any) used for the analysis, results and policy implications and recommendations. The final paper should be no longer than eight pages of A4 format including references, charts and tables. This paper, counted as 70% of the final group project grade, is due by COB on the last day of class (April 28). *All students in each project group will receive the same grade for the entire project.*

Late work: Late papers will receive:

- a 10% deduction if submitted within 1 day of the due date;
- 25% deduction if submitted within 1-2 days of the due date;
- 50% deduction if submitted within 2-3 days of the due date.
- No late work will be accepted after this time.

Midterm and Final Exam (20 and 39% respectively): The midterm will be held in the last class before the Spring recess and the final exam will be held on May XX.

Course Schedule (see below for readings)

Week	Topic	Assessment
1	Course introduction and background	
2	Electricity Markets 1: Markets Background and Regulation	
3	Electricity Markets 2: Deregulation and Competition	
4	Electricity Markets 3: Reliability	Quiz 1
5	Empirical Methods in Energy Economics	
6	Empirical Methods in Energy Economics Cont. Finite Fossil Fuels	
7	Finite Fossil Fuels Cont.	
8	Renewables	Midterm (60 mins)
9	SPRING BREAK	
10	Climate Externalities and Policy	
11	Innovation Externalities, Learning by Doing	Project Proposals Due, Quiz 2
12	Energy Demand Flexibility, Transportation	
13	Welfare Economics and Energy Economics	Quiz 3
14	Group Presentations	
15	Group Presentations, Review Session	Project Paper Due (April 28)
16	Study Break	Final Exam (120 mins) (TBA)

Suggested textbooks to help:

Filippini, Massimo, and Suchita Srinivasan. [*"An Introduction to Energy Economics and Policy."*](#) Cambridge University Press, 2024.

Schwarz, Peter. [*"Energy Economics"*](#); New York, Routledge, 2018.

Reading List

Weeks 2--4 electricity markets: regulation, deregulation, liberalization and economic dispatch, reliability, value of lost load

- Averch, Harvey, and Leland L. Johnson. ["Behavior of the firm under regulatory constraint."](#) *The American Economic Review* 52.5 (1962): 1052-1069.
- Borenstein, Severin, and James B. Bushnell. ["Do two electricity pricing wrongs make a right? Cost recovery, externalities, and efficiency."](#) *American Economic Journal: Economic Policy* 14, no. 4 (2022): 80-110
- Joskow, P.L. (2008) ["Lessons learned from electricity market liberalization"](#), *Energy Journal* vol. 29, Special Issue no. 2
- Borenstein, Severin, and James Bushnell. ["The US electricity industry after 20 years of restructuring."](#) *Annu. Rev. Econ.* 7, no. 1 (2015): 437-463.
- Cicala, Steve. ["Imperfect markets versus imperfect regulation in US electricity generation."](#) *American Economic Review* 112.2 (2022): 409-441.
- Fabrizio, K., Rose, N. Wolfram, C. (2007), ["Do Markets Reduce Costs? Assessing the Impact of Regulatory Restructuring on U.S. Electric Generation Efficiency,"](#) *American Economic Review*, Vol. 97: 1250-1277.
- Borenstein, Severin, James Bushnell, and Erin Mansur. ["The economics of electricity reliability."](#) *Journal of Economic Perspectives* 37, no. 4 (2023): 181-206.
- Brown, David P., and Lucija Muehlenbachs. ["The value of electricity reliability: Evidence from battery adoption."](#) *Journal of Public Economics* 239 (2024): 105216.
- Gorman, Will. ["The quest to quantify the value of lost load: A critical review of the economics of power outages."](#) *The Electricity Journal* 35.8 (2022): 107187.

Weeks 5 & 6: Empirical Methods in Energy Economics

- Angrist, J. D. & Pischke, J.-S. (2014). ["Mastering 'metrics: The path from cause to effect."](#) Princeton University Press, 2014.
- Hahn, Robert, and Robert Metcalfe. ["The impact of behavioral science experiments on energy policy."](#) *Economics of Energy & Environmental Policy* 5, no. 2 (2016): 27-44.
- Hahn, Robert W., and Robert D. Metcalfe. ["Efficiency and equity impacts of energy subsidies."](#) *American Economic Review* 111, no. 5 (2021): 1658-1688.
- Brandon, Alec, John A. List, Robert D. Metcalfe, Michael K. Price, and Florian Rundhammer. ["Testing for crowd out in social nudges: Evidence from a natural field experiment in the market for electricity."](#) *Proceedings of the National Academy of Sciences* 116, no. 12 (2019): 5293-5298.
- Jessoe, Katrina, and David Rapson. ["Knowledge is \(less\) power: Experimental evidence from residential energy use."](#) *American Economic Review* 104, no. 4 (2014): 1417-1438.
- Borenstein, Severin, and James Bushnell. ["The US electricity industry after 20 years of restructuring."](#) *Annu. Rev. Econ.* 7, no. 1 (2015): 437-463.

Weeks 6 & 7: Finite Fossil Fuels

- Thomas Covert, Michael Greenstone, and Christopher R Knittel. ["Will we ever stop using fossil fuels?"](#) *The Journal of Economic Perspectives*, 30(1): 117-137, 2016

- Kilian, Lutz. 2016. "[The Impact of the Shale Oil Revolution on U.S. Oil and Gasoline Prices.](#)" *Review of Environmental Economics and Policy* 10 (2): 185–205.
- Livernois, John. 2009. "[On the Empirical Significance of the Hotelling Rule.](#)" *Review of Environmental Economics and Policy* 3 (1): 22–41.
- Muehlenbachs, L., Spiller, E., & Timmins, C. (2015). [The housing market impacts of shale gas development.](#) *American Economic Review*, 105(12), 3633–3659.
- Bartik, A. W., Currie, J., Greenstone, M., & Knittel, C. R. (2019). [The local economic and welfare consequences of hydraulic fracturing.](#) *American Economic Journal: Applied Economics*, 11(4), 105–155.
- Abuin, C. (2024). [Power Decarbonization in a Global Energy Market: The Climate Effect of US LNG Exports.](#) Working paper.
- Kellogg, R. (2024). [The End of Oil](#) (No. w33207). National Bureau of Economic Research.
- Acemoglu, Daron, Philippe Aghion, Lint Barrage, and David Hémous (2024) "[Climate Change, Directed Innovation, and Energy Transition: The Long-run Consequences of the Shale Gas Revolution](#)" NBER WP No. 31657

Week 9 – The Economics of Renewables

- Borenstein, S. (2012). [The private and public economics of renewable electricity generation.](#) *Journal of Economic Perspectives*, 26(1), 67–92.
- Breyer et al. (2022). [On the history and future of 100% renewable energy systems research.](#) *IEEE Access*, 10, 78176–78218.
- Davis, L. W., Hausman, C., & Rose, N. L. (2023). [Transmission impossible? Prospects for decarbonizing the US grid.](#) *Journal of Economic Perspectives*, 37(4), 155–180.
- Schmalensee, Richard. 2012. "[Evaluating Policies to Increase Electricity Generation from Renewable Energy.](#)" *Review of Environmental Economics and Policy* 6 (1): 45–64.

Week 10 - Climate Externalities and Policy

- Hsiang, S., & Kopp, R. E. (2018). [An economist's guide to climate change science.](#) *Journal of Economic Perspectives*, 32(4), 3–32.
- Carleton, Tamma, and Michael Greenstone (2021). "[Updating the United States government's social cost of carbon.](#)" *University of Chicago, Becker Friedman Institute for Economics Working Paper* 2021-04 (2021).
- Parry, I., Black, M. S., & Vernon, N. (2021). [Still not getting energy prices right: a global and country update of fossil fuel subsidies.](#) International Monetary Fund.
- Rennert, Kevin, Frank Errickson, Brian C. Prest, Lisa Rennels, Richard G. Newell, William Pizer, Cora Kingdon, et al. 2022. "[Comprehensive Evidence Implies a Higher Social Cost of CO₂.](#)" *Nature* 610 (7933): 687–92.
- Carleton, Tamma A., and Solomon M. Hsiang. 2016. "[Social and Economic Impacts of Climate.](#)" *Science*, 353 (6304)
- Auffhammer, Maximilian. 2018. "[Quantifying Economic Damages from Climate Change.](#)" *Journal of Economic Perspectives* 32 (4): 33–52.

Week 11 - Innovation Externalities

- Dugoua, E., & Moscona, J. (2025). [The Economics of climate innovation: technology, climate policy, and the clean energy transition](#) (No. w34601). National Bureau of Economic Research.
- Nemet, Gregory F. [How solar energy became cheap: A model for low-carbon innovation.](#) Routledge, 2019.
- Aghion, P., Hepburn, C., Teytelboym, A., & Zenghelis, D. (2019). [Path dependence, innovation and the economics of climate change.](#) *Handbook on green growth*, 67–83.

- Aghion, P., Dechezleprêtre, A., Hemous, D., Martin, R., & Van Reenen, J. (2016). [Carbon taxes, path dependency, and directed technical change: Evidence from the auto industry](#). *Journal of Political Economy*, 124(1), 1-51.

Week 11 – Energy Demand Flexibility and Transportation

Demand Flexibility

- Bernard, L., Hackett, A., Metcalfe, R. D., & Schein, A. (2024). [Decarbonizing Heat: The Impact of Heat Pumps and a Time-of-Use Heat Pump Tariff on Energy Demand](#) (No. w33036). National Bureau of Economic Research.
- Metcalfe, R. D., Schein, A. R., Simpson, C. R., & Sun, Y. (2025). [AI in charge: Large-scale experimental evidence on electric vehicle charging demand](#). Technical report, Working paper.

Transportation

- Bernard, L., Hackett, A., Metcalfe, R. D., Panzone, L., & Schein, A. (2025). [The impact of dynamic prices on electric vehicle public charging demand: Evidence from a nationwide natural field experiment](#) (No. w34600). National Bureau of Economic Research.
- Rapson, D., & Muehlegger, E. (2023). [Global transportation decarbonization](#). *Journal of Economic Perspectives*, 37(3), 163-188.
- Rapson and Muehlegger (2022) [Economics of Electric Vehicles](#). *Review of Environmental Economics and Policy*.

Week 12 – Guest lecture and the Economics of Net Zero Energy Policy.

- IEA (2021) [Net Zero by 2050: A Roadmap for the Global Energy Sector](#).
- DeAngelo, J., Azevedo, I., Bistline, J., Clarke, L., Luderer, G., Byers, E., & Davis, S. J. (2021). [Energy systems in scenarios at net-zero CO₂ emissions](#). *Nature communications*, 12(1), 6096.
- Fankhauser, S., Smith, S. M., Allen, M., Axelsson, K., Hale, T., Hepburn, C., ... & Wetzter, T. (2022). [The meaning of net zero and how to get it right](#). *Nature Climate Change*, 12(1), 15-21.

Week 13 – Welfare Economics and Energy Economics

- Hahn, Robert W., Nathaniel Hendren, Robert D. Metcalfe, and Ben Sprung-Keyser. [A welfare analysis of policies impacting climate change](#). No. w32728. National Bureau of Economic Research, 2024.
- Finkelstein, Amy, and Nathaniel Hendren. "[Welfare analysis meets causal inference](#)." *Journal of Economic Perspectives* 34, no. 4 (2020): 146-167.

Academic Integrity

Columbia's intellectual community relies on academic integrity and responsibility as the cornerstone of its work. Graduate students are expected to exhibit the highest level of personal and academic honesty as they engage in scholarly discourse and research. In practical terms, you must be responsible for the full and accurate attribution of the ideas of others in all of your research papers and projects; you must be honest when taking your examinations; you must always submit your own work and not that of another student, scholar, or internet source. Graduate students are responsible for knowing and correctly utilizing referencing and bibliographical guidelines. When in doubt, consult your professor.

Failure to observe these rules of conduct will have serious academic consequences, up to and including dismissal from the university. If a faculty member suspects a breach of academic honesty, appropriate investigative and disciplinary action will be taken.

